Tile cache replay study

Method: 200 recorded camera paths were replayed against caches of 128, 256, 512 and 1024 tiles.

Hit rate with least recently used eviction:

128 tiles 71 percent

256 tiles 84 percent

512 tiles 95 percent

1024 tiles 97 percent

Eviction policies at 256 tiles
Least recently used 84 percent
First in, first out 77 percent
Random replacement 73 percent
Conclusion: keep least recently used with a pinned viewport set.
Above 512 tiles extra capacity adds under 2 points.